

Heart Failure 30-Day Readmission Prediction

Project Analysis

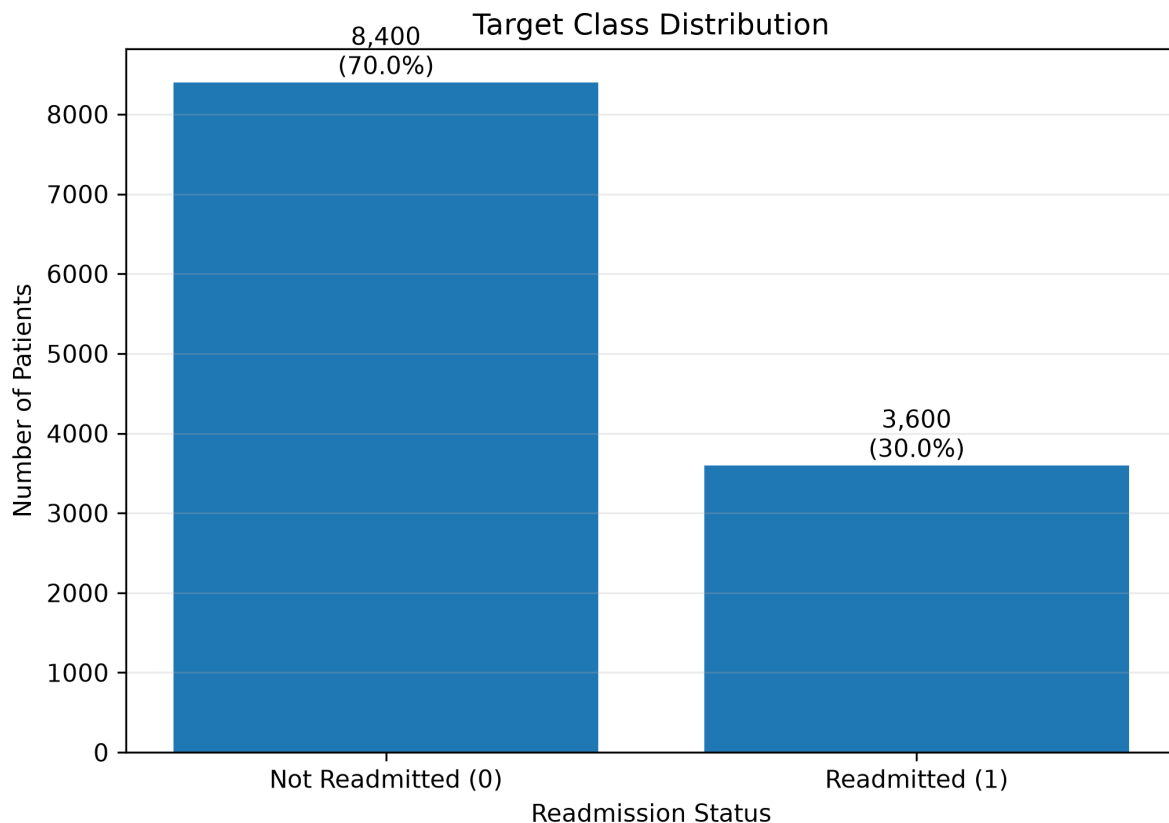
Dataset: 12,000 patient records | Models: Logistic Regression, KNN, Decision Tree

1. Executive Summary

The dataset contains 12,000 patient records and 36 columns, including a binary target named `Readmitted_30_Days`. The target contains 8,400 non-readmitted patients (70%) and 3,600 readmitted patients (30%), so the data are moderately imbalanced. `Patient_ID` was excluded because it is an identifier with one unique value per record.

A stratified 80/20 train-test split with `random_state=42` was used. Missing-value handling, one-hot encoding, and scaling were placed inside scikit-learn pipelines so preprocessing parameters were learned from training data only. Five-fold stratified cross-validation on the training set was used for hyperparameter selection.

The tuned Logistic Regression model was the strongest overall model: test accuracy 0.895, precision 0.852, recall 0.786, F1-score 0.818, and ROC-AUC 0.956. Its train/test metrics were also extremely close, providing strong evidence of good generalization rather than overfitting.



2. Dataset Exploration and Data Quality

Shape: 12,000 rows $\times$ 36 columns.

Missing values: 0 across the complete dataset.

Duplicate rows: 0.

Target: Readmitted_30_Days, encoded as 0 = not readmitted within 30 days and 1 = readmitted within 30 days.

Class distribution: 0 = 8,400 (70.0%), 1 = 3,600 (30.0%).

Categorical predictors are Gender, Smoking_Status, Alcohol_Consumption, and Heart_Failure_Type. Patient_ID has 12,000 unique values and is treated as an identifier and excluded.

3. Leakage Assessment

No column directly encodes the target except Readmitted_30_Days itself, which was removed from X. Patient_ID was also removed. Variables such as Length_of_Stay, ICU_Admission, Emergency_Admission, and medication indicators may be legitimate predictors only if known at the intended prediction time. If prediction is intended earlier than discharge, any variable recorded afterward would constitute temporal leakage. The supplied CSV does not provide enough timestamp/data-dictionary information to establish that distinction, so this remains a deployment-level limitation.

4. Preprocessing and Experimental Design

The same stratified 80/20 split was used for all three models. Numeric features were median-imputed and standardized for Logistic Regression and KNN. Categorical features were most-frequent imputed and one-hot encoded. Decision Tree used the same imputation/encoding approach but did not require numeric scaling.

The preprocessing operations are embedded inside pipelines, so imputation statistics, scaling parameters, and category mappings are learned from training data rather than the test set.

Hyperparameters were selected with 5-fold stratified cross-validation on the training set using ROC-AUC as the selection criterion. The final test set was kept separate.

5. Selected Hyperparameters

Logistic Regression: $C = 10$.

KNN: $k = 15$.

Decision Tree: $\text{max_depth} = 7$, $\text{min_samples_leaf} = 10$.

6.Overfitting and Underfitting Analysis

1. Logistic Regression — Good Generalization

Metric	Training	Testing	Gap
Accuracy	0.894	0.895	0.001
ROC-AUC	0.954	0.956	0.002

Analysis:

Logistic Regression does not show significant overfitting. Its training and testing performance are almost identical. In fact, the test accuracy and ROC-AUC are marginally higher than the training values, which can occur because the test set is a different finite sample. This indicates that the model has learned patterns that generalize well to unseen data.

Diagnosis: GOOD GENERALIZATION

It also does not show evidence of underfitting because both training and testing performance are strong rather than simultaneously low.

2. KNN — Mild Overfitting

Metric	Training	Testing	Gap
Accuracy	0.865	0.851	0.014
ROC-AUC	0.947	0.918	0.029

Analysis:

KNN performs better on the training data than on the test data. The ROC-AUC decreases from 0.947 to 0.918, giving a gap of approximately 0.029.

This indicates that KNN has learned some patterns that are specific to the training data and does not generalize quite as well to unseen patients.

However, the gap is not extremely large, so the evidence supports mild rather than severe overfitting.

Diagnosis: MILD OVERFITTING

3. Decision Tree — Moderate Overfitting

Metric	Training	Testing	Gap
Accuracy	0.875	0.839	0.036
ROC-AUC	0.931	0.895	0.036

Analysis:

The Decision Tree shows the largest generalization gap among the three models.

Its accuracy falls from 87.5% on training data to 83.9% on test data, a gap of 3.6 percentage points.

Similarly, ROC-AUC falls from 0.931 to 0.895, also a 0.036 gap.

This provides clear evidence that the tree is fitting the training data more closely than it can generalize to unseen patients. The tree therefore shows moderate overfitting.

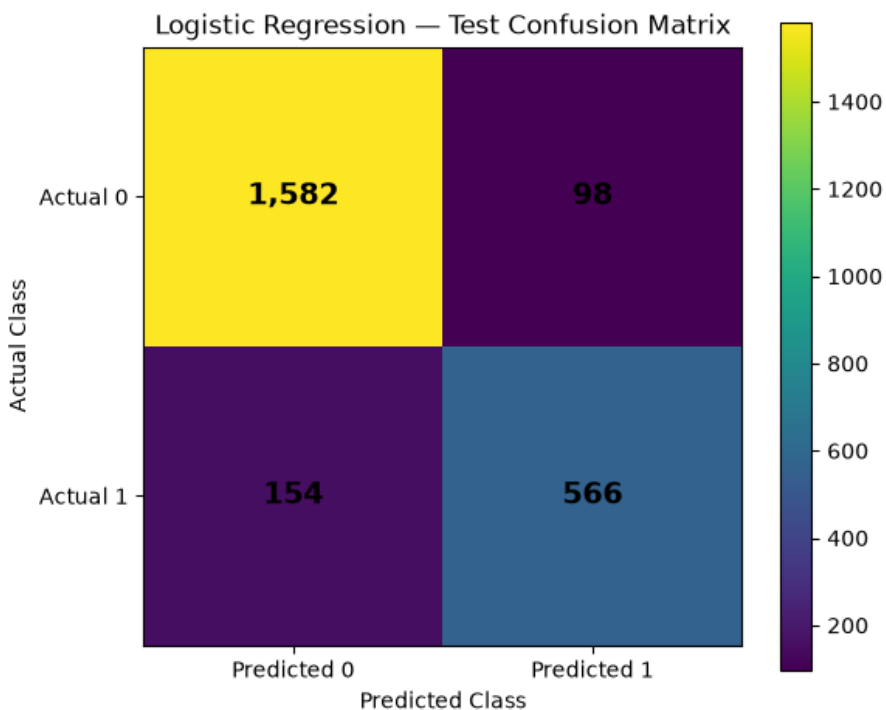
Diagnosis: MODERATE OVERFITTING

7. Mandatory Model Comparison

Model	Train Acc.	Test Acc.	Precision	Recall	F1	ROC-AUC	Fit
Logistic Regression	0.894	0.895	0.852	0.786	0.818	0.956	Good generalization
KNN	0.865	0.851	0.833	0.588	0.703	0.918	Mild overfitting
Decision Tree	0.875	0.839	0.753	0.690	0.720	0.895	Moderate overfitting

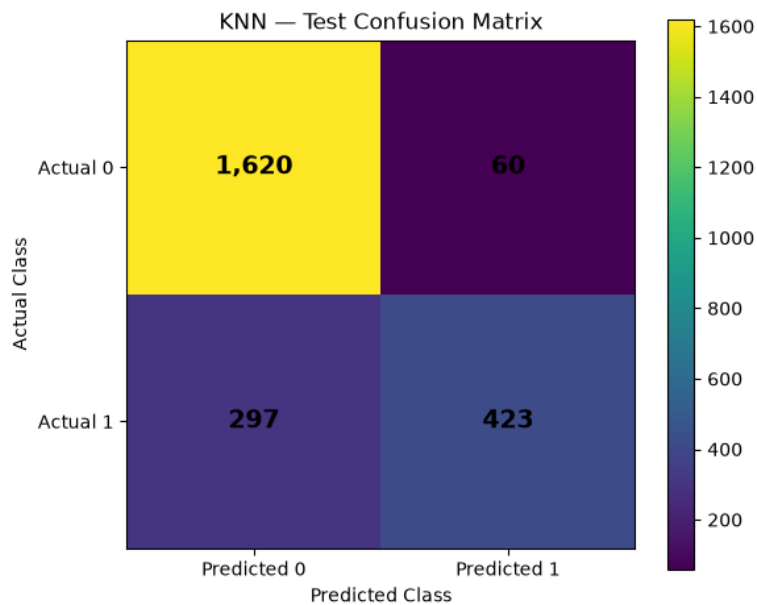
Logistic Regression is the strongest overall model because it leads on every required test metric and has the smallest train-test performance gap.

8. Logistic Regression — Confusion Matrix



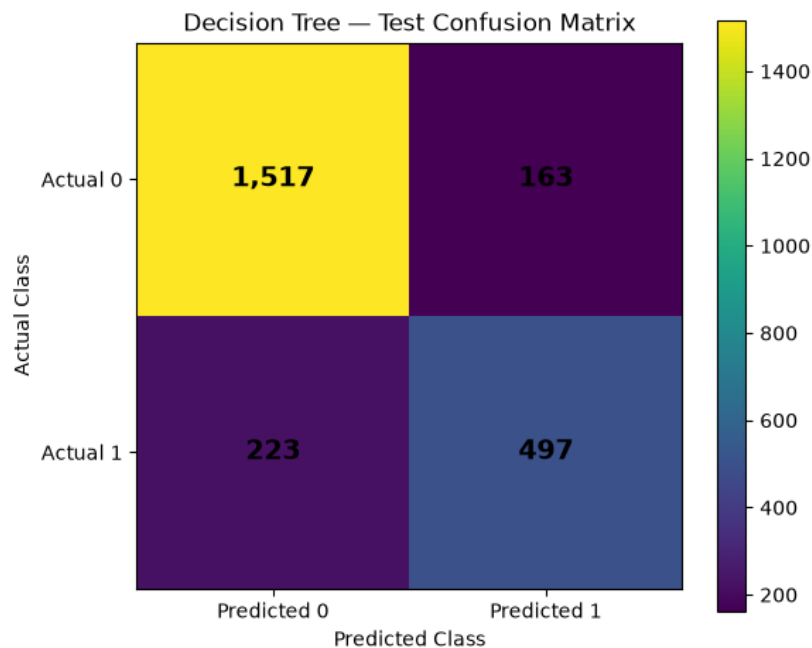
TN is a correctly identified non-readmission, FP is a patient predicted as readmitted who was not, FN is a patient actually readmitted but missed, and TP is a correctly identified readmission.

9. KNN — Confusion Matrix



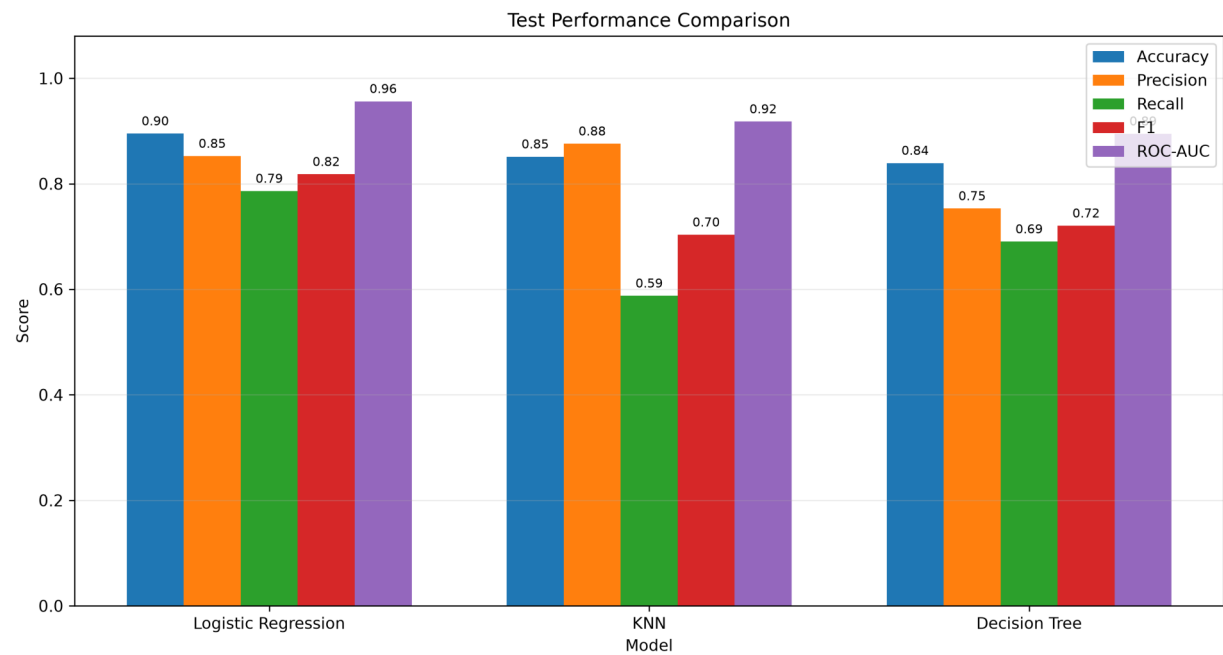
TN is a correctly identified non-readmission, FP is a patient predicted as readmitted who was not, FN is a patient actually readmitted but missed, and TP is a correctly identified readmission.

10. Decision Tree — Confusion Matrix

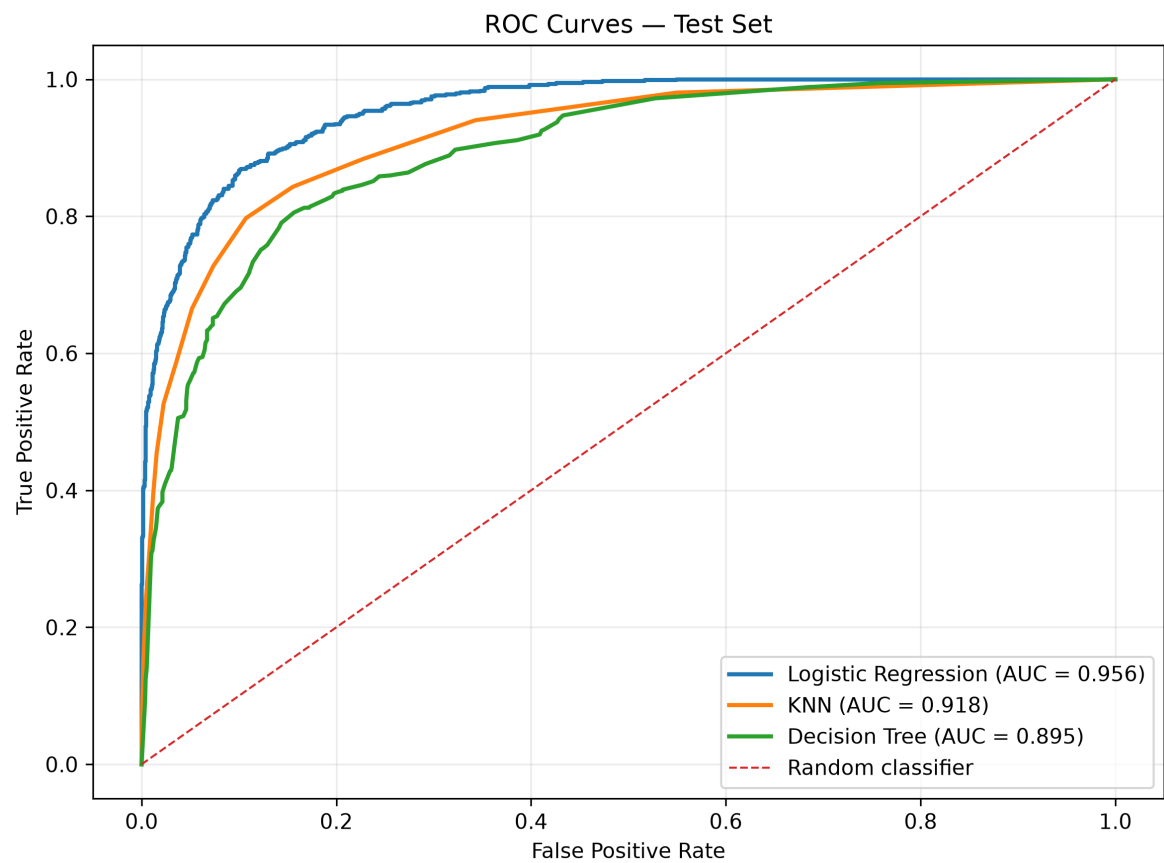


TN is a correctly identified non-readmission, FP is a patient predicted as readmitted who was not, FN is a patient actually readmitted but missed, and TP is a correctly identified readmission.

11. Metric Comparison

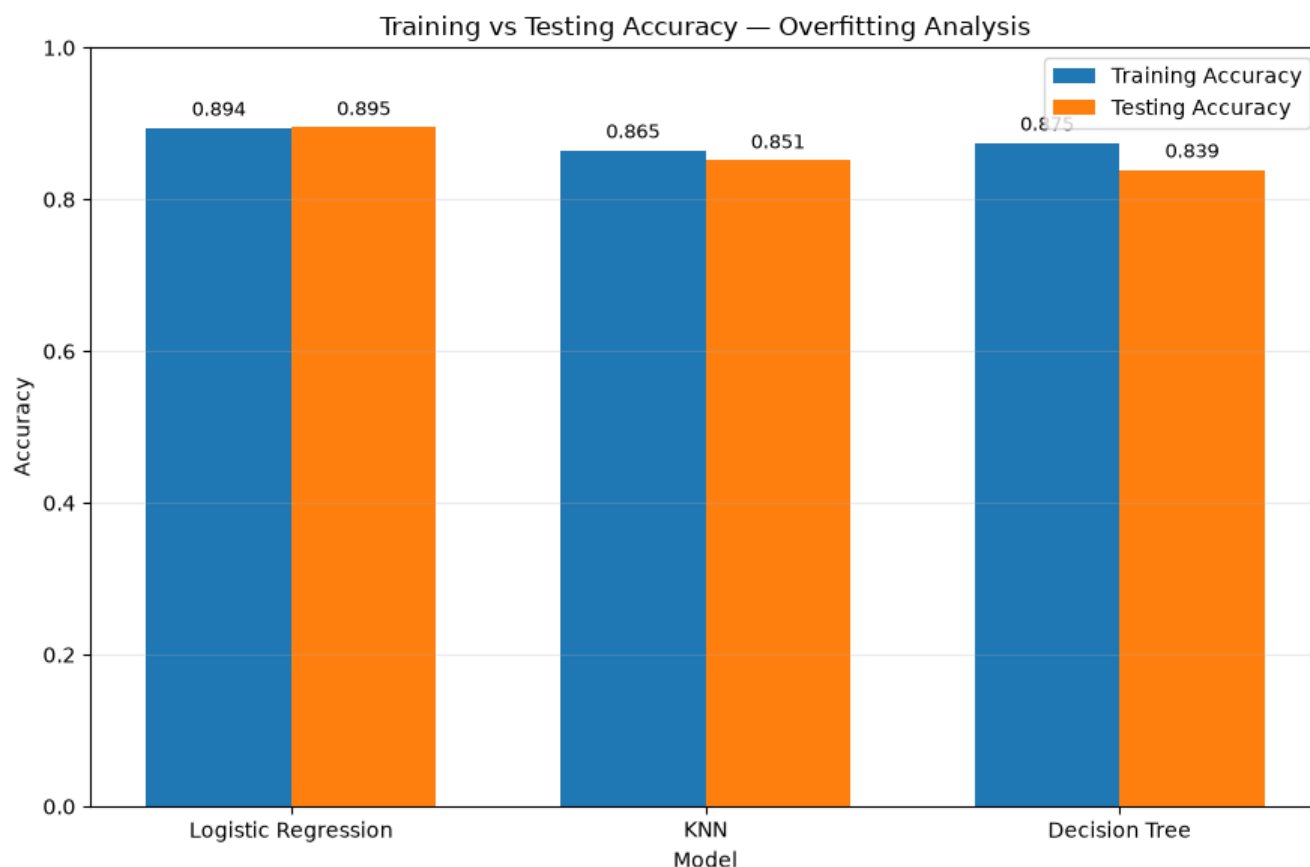


12. ROC Curves



Logistic Regression has the highest test ROC-AUC (0.956), followed by KNN (0.918) and Decision Tree (0.895).

13. Training vs Test Performance



14. Answers to Q1–Q14

Q1. What is the class distribution? Is it balanced?

The target is 70% class 0 (8,400 patients) and 30% class 1 (3,600 patients). It is moderately imbalanced. Accuracy alone can hide poor minority-class performance, so recall, precision, F1 and ROC-AUC should also be examined.

Q2. What preprocessing was performed?

Patient_ID was removed as an identifier. Categorical variables were one-hot encoded. Numeric variables were median-imputed and standardized for Logistic Regression and KNN; the tree used imputation and encoding without scaling. These operations were implemented inside pipelines and fitted using training data only.

Q3. Why is scaling much more important for KNN than Logistic Regression?

KNN calculates distances, so a feature with a larger numerical scale can dominate unless features are standardized. Logistic Regression is coefficient-based rather than distance-based, although scaling remains useful for numerical conditioning and regularization.

Q4. What k was selected for KNN and why?

k=15 was selected by 5-fold cross-validation using ROC-AUC. The tested values were 3, 5, 7, 11 and 15. Small k values are more sensitive to noise and have higher variance; very large k values can oversmooth the decision boundary and increase bias.

Q5. What Decision Tree settings were used?

The selected tree used max_depth=7 and min_samples_leaf=10. These constraints limit tree complexity and reduce highly specific terminal regions, improving generalization.

Q6. Compare the models and identify the best.

Logistic Regression achieved test Accuracy=0.895, Precision=0.852, Recall=0.786, F1=0.818 and ROC-AUC=0.956. KNN achieved 0.851, 0.833, 0.588, 0.703 and 0.918. Decision Tree achieved 0.839, 0.753, 0.690, 0.720 and 0.895. Logistic Regression is the strongest overall choice because it leads on every listed test metric.

Q7. Which metric deserves most attention in healthcare?

Recall deserves particularly strong attention when false negatives are serious because a missed high-risk patient may fail to receive timely follow-up. Precision also matters because excessive false positives consume clinical resources. The final threshold should therefore reflect the hospital's relative FN and FP costs.

Q8. Does any model appear overfit?

KNN shows train accuracy 0.865 versus test 0.851 and train ROC-AUC 0.947 versus test 0.918. Decision Tree shows train accuracy 0.875 versus test 0.839 and train ROC-AUC 0.931 versus test 0.895. These gaps support mild KNN overfitting and more noticeable tree overfitting. Logistic Regression has virtually no accuracy or ROC-AUC gap.

Q9. Does any model appear underfit?

There is no strong evidence of underfitting. Underfitting normally shows low training and low test performance. Logistic Regression has high, closely matched train/test performance, while KNN and the tree show higher training than test performance.

Q10. How could overfitting be reduced?

For KNN, increase k if validation confirms benefit. For the Decision Tree, reduce max_depth, increase min_samples_leaf/min_samples_split, or use pruning. Cross-validation, justified feature selection, and more diverse training data can also reduce variance.

Q11. How could underfitting be improved?

If future experiments show underfitting, increase model capacity, reduce excessive regularization, engineer informative features, and improve preprocessing. For Logistic Regression, clinically justified

nonlinear features or interactions could add representation capacity. Current results do not strongly indicate underfitting.

Q12. Is the highest accuracy necessarily the best model?

No. Accuracy alone is insufficient. KNN has 85.1% accuracy but only 58.8% recall, meaning it misses a comparatively large fraction of actual readmissions. Healthcare model selection must consider false-positive and false-negative consequences.

Q13. If two models have similar ROC-AUC but different recall/precision, which would you choose?

Choose based on the clinical objective and relative cost of false negatives versus false positives. If identifying as many true readmissions as possible is the priority, favor higher recall if the false-positive burden is acceptable. If unnecessary alerts are costly, higher precision may be preferable. The probability threshold can be adjusted without retraining.

Q14. What are limitations before real-world deployment?

The dataset does not establish temporal availability of every feature, so variables recorded after the intended prediction time could cause leakage. External validation on another hospital/time period is required. Dataset shift and prevalence changes can alter performance. Fairness across demographic groups should be assessed. Calibration, clinical-utility analysis, threshold selection, and prospective validation are needed before clinical deployment. The model should support, not replace, clinical judgment.

15. Final Verdict

The selected Logistic Regression model is the best-performing model in this experiment. It achieved the highest test Accuracy (89.5%), Precision (85.2%), Recall (78.6%), F1-score (81.8%), and ROC-AUC (95.6%). Its training and test performance are nearly identical, providing strong evidence of good generalization on this held-out split.

For a healthcare use case, the implementation should not simply deploy a 0.5 probability threshold. The threshold should be selected using the hospital's acceptable false-negative and false-positive costs, followed by calibration and external/prospective validation. This is a project-level model-selection result, not evidence that the model is ready for clinical deployment.